

Lung Nodule Segmentation — SegResNet 3D — small

1. Task

Voxel-level segmentation of pulmonary nodules in 3D chest CT volumes, cropped to a per-series lung bounding box.

- **Input** (1, 256, 256, 256) CT crop, intensity-normalised to [0, 1], resampled from the bbox (padding = 20 vox).
- **Output** (2, 256, 256, 256) softmax logits — class 0 = background, class 1 = nodule.

2. Model

[MONAI SegResNet](#), SegResNet (residual U-Net).

Property	Value
Architecture family	SegResNet (residual U-Net)
Spatial dims	3
Input channels	1
Output channels	2 (softmax)
Initial feature width	16 (small)
Encoder blocks per level	[1, 2, 2, 4, 4]
Decoder blocks per level	[1, 1, 1, 1]
Dropout	0.1
Trainable parameters	20,663,538

3. Data

Trained on the **unified** split (patient-grouped, dataset-stratified, **full corpus**: NLST + NSCLC + LIDC-IDRI).

Split sizes: **1 683 train / 297 val / 325 test (held out)**.

Class imbalance: nodule voxels are on the order of $\sim 10^{-5}$ of the total, which shapes how each metric should be read (§7).

4. Training

Setting	Value
Loss	Focal Tversky + weighted CE ($\alpha=0.3$, $\beta=0.7$, $\gamma=2.0$, $\lambda_{CE}=0.1-0.3$, <code>ce_nodule_weight</code> = 100)
Optimizer	Adam
Learning rate	1×10^{-5}
Weight decay	1×10^{-5}
LR schedule	Cosine annealing, $T_{max} = 400$, $\eta_{min} = 1 \times 10^{-6}$
Batch size	4
Epochs	400
Random seed	42
Mixed precision	bf16 (autocast, no GradScaler)
Augmentation	3D flips, 90° rotations, elastic rotation, zoom, intensity scale/shift, Gaussian noise/blur, contrast
Hardware	1 × NVIDIA H100 94 GB
Wall-clock	≈ 3.5 days

Best checkpoint (evaluated below): **epoch 339 / 400**.

5. Evaluation protocol

Held-out test split (325 series). The training loop used the `val` split of `unified` for early stopping and best-Dice checkpoint selection; the numbers below come from the completely untouched `test` split (never seen during training or model selection). Predictions taken as `argmax` over the 2-channel softmax output; foreground = class 1. All metrics are computed at the voxel level and micro-averaged over the whole test split (5,452,595,200 voxels total).

6. Results

Ticket-required metrics

Metric	Value
mean IoU	0.7391
Accuracy	0.9991
Precision	0.5389
Recall	0.8119

Supplementary

Metric	Value
IoU (foreground / nodule class)	0.4791
IoU (background class)	0.9991
Dice / F1 (micro)	0.6478

Per-case Dice distribution (325 val cases)

Statistic	Value
Mean	0.5240
Std	0.2726
Min	0.0000
p05	0.0024
p25	0.3158
Median	0.5846
p75	0.7521
p95	0.8788
Max	0.9528

7. Notes on interpretation

Nodule segmentation is severely class-imbalanced ($\sim 10^{-5}$ of voxels are nodule), which distorts the standard metric set:

- **Accuracy** is trivially near 1.0 for any reasonable model — the model gets $\approx 99.9133\%$ of voxels right by predicting "background" almost everywhere. Uninformative on its own here.
- **mIoU** averages the foreground and background IoU. `IoU_background` is essentially 1.0, so mIoU is roughly $0.5 + 0.5 \cdot \text{IoU_nodule}$. The useful signal is in `IoU_nodule` (0.4791) and Dice / F1 (0.6478).
- **Precision / Recall** are the standard per-voxel figures — no imbalance caveat needed. Recall 0.8119 means the model correctly labels $\sim 81\%$ of nodule voxels; Precision 0.5389 means $\sim 54\%$ of predicted-nodule voxels are true nodule.

Per-case Dice mean (0.5240) is lower than the micro Dice (0.6478) because large nodules dominate the micro average; per-case Dice weights each patient equally regardless of nodule size. Clinically the per-case distribution is the more useful summary.

8. How to load & run inference

Pre-trained weights are hosted at [HalmosiL/nodule-segresnet-3d-small](#). Download and load directly:

```
huggingface-cli download HalmosiL/nodule-segresnet-3d-small --local-dir ./ckpt
```

```
import yaml, torch
from monai.networks.nets import SegResNet

cfg = yaml.safe_load(open("config.yaml"))["model"]
model = SegResNet(
    spatial_dims = cfg["spatial_dims"],
    in_channels   = cfg["in_channels"],
    out_channels  = cfg["out_channels"],          # 2
    init_filters  = cfg["init_filters"],
    blocks_down   = tuple(cfg["blocks_down"]),
    blocks_up     = tuple(cfg["blocks_up"]),
    dropout_prob  = cfg["dropout_prob"],
)
state = torch.load("model.pth", map_location="cpu", weights_only=True)
model.load_state_dict(state)
model.eval()

with torch.no_grad():
    # Input: (B, 1, 256, 256, 256), lung-bbox-cropped CT resampled to 2563.
    x = torch.randn(1, 1, 256, 256, 256)
    logits = model(x)                                # (B, 2, D, H, W)
    pred_class = logits.argmax(dim=1)                  # (B, D, H, W) in {0, 1}
    nodule_mask = (pred_class == 1).to(torch.uint8)
```

This model expects **lung-bbox-cropped** input. A separate 2D ROI model is needed to produce that bbox — see the accompanying ROI checkpoints ([HalmosiL/roi-segresnet-2d](#) or [HalmosiL/roi-swinunetr-2d](#)), or use the demo's `inference.py` which chains them for you.

Instance-level metrics (per-nodule)

Connected-component analysis with 26-connectivity, pooled over the held-out test split, on the 256-cube resample grid (same grid as the voxel metrics above).

- **Instance recall** — GT nodules touched by at least one predicted voxel, over all GT nodules.
- **Instance precision** — predicted components touching at least one GT voxel, over all predicted components.

Metric	Value
GT nodules (components)	941

Metric	Value
GT nodules hit	740
Instance recall	0.7864
Predicted components	2,025
Predicted components hitting GT	748
Instance precision	0.3694

No minimum-size filtering is applied — every predicted component counts, so single-voxel false positives lower instance precision. Produced by `scripts/instance_metrics.py --config configs/segresnet_small.yaml --ckpt <ckpt> --split test` (JSON: `results_instance/segresnet_small.json`).

9. Reproducibility

- Config `configs/segresnet_small.yaml`
- Checkpoint `checkpoints/segresnet_small/best_model.pth`
- Metrics JSON `checkpoints/segresnet_small/eval_metrics_nodule_test.json`
- Command `python eval_metrics_nodule.py --config configs/segresnet_small.yaml --ckpt checkpoints/segresnet_small/best_model.pth --split test`

Companion two-stage nodule models on HuggingFace:

- [HalmosiL/nodule-segresnet-3d-wide](#) — wider SegResNet (resize recipe)
- [HalmosiL/nodule-dynunet-3d](#) — DynUNet (3D U-Net)